SECTION & DOCUMENT LIST - LOCK ASCII

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                        LIST LOCK 🔒

📂 P.B.F. EXPANTION
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META-ABSTRACT

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This document forms part of **C. P.B.F. EXPANSION**, within **C. VISUAL EXPANTION P.B.F**, and occupies the visual-representation position within the P.B.F. expansion material.

Its function is to **PROVIDE** the collected visual presentation through which the established P.B.F. structure is represented across linear, bidirectional, planar, quadratic, cubic, scale-based, and Rx8 relational reference fields. The visual sequence makes the continuity of the established operational identity inspectable as direction, dimension, field size, grouping, scale, and representation change.

Within the complete expansion, the document relates the established P.B.F. structure to the transformation and projection material presented in S6 and to the multiple-reading distinction established in S7. It provides the visual field through which dimensional expansion, center-oriented execution, self-similar extension, simultaneous relational readings, and the Rx8 reference architecture can be inspected as different representations of preserved relational structure.

The document therefore provides the visual and geometric representations through which these downstream relations are displayed, compared, and read across multiple reference fields. It does not make the multiplication of directions, dimensions, scales, geometric

fields, or simultaneous readings equivalent to the generation of new primary functions, and it does not make a later representational framework the origin of relations established upstream.

This meta-abstract describes the document's functional position in the expansion material. It does not limit the material that must be read or treated as relevant during review.

ⓘ DOCUMENT IDENTITY

- **REPOSITORY: *RP9. RH. Z0***
  - **RP9. EXTRA ACCESSORIES** (RH. Z0)

- SUPPORT SOURCE: *RP9 Riemann Submission*
  - SUPPORT SOURCE: **REPOSITORY:** *RP9. RH. Z0*

Section: ***C. P.B.F. EXPANSION***
Subsection: ***C. VISUAL EXPANTION P.B.F***
LETTER: ***C***
Official title: ***Images SVG Presentation***

Visual Expansion P.B.F.

- README — VISUAL EXPANSION P.B.F
- **1A. LINEAR DIRECTIONAL P.B.F. svg**



The README document establishes:

- the separation of the accessory from the main repository;
- the eight-part `1A–4B` structure;
- the distinction between established function and geometric representation;
- shared colors, symbols, and SVG rules;
- the varying authority of the source materials;
- that legacy S6 material may only be used where compatible with the current P.B.F. structure.

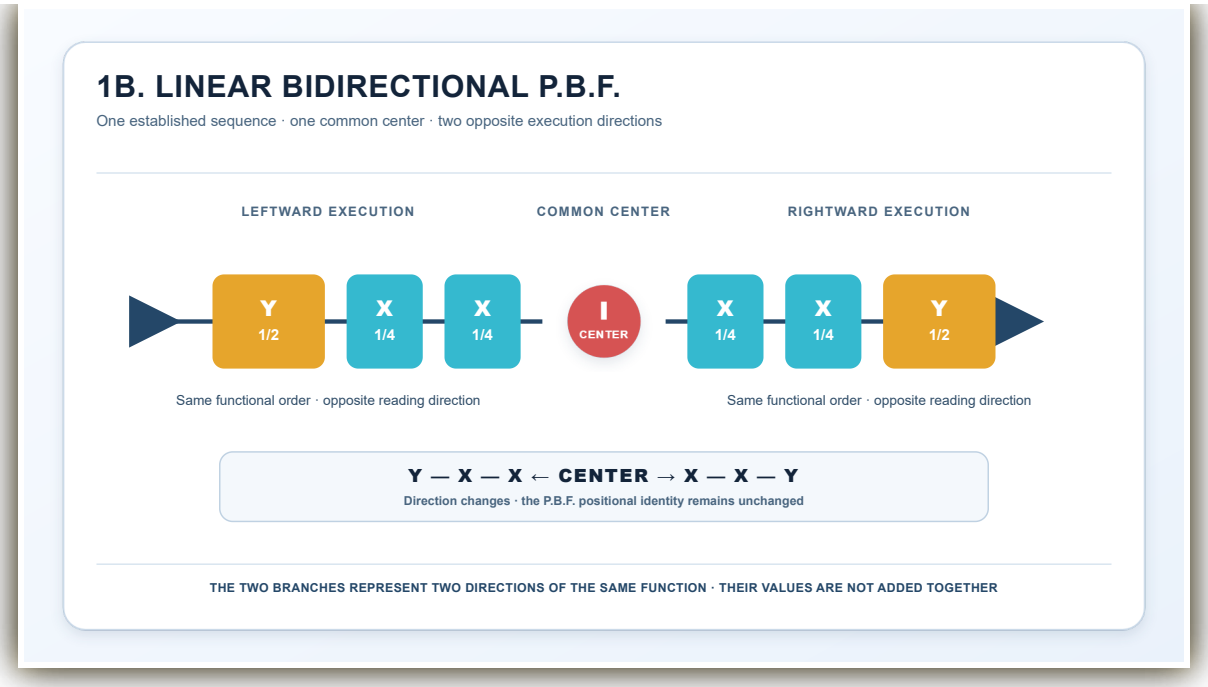
The first SVG image shows exactly:

$$I \longrightarrow \frac{I}{2} + \frac{I}{2} \longrightarrow \frac{I}{4} + \frac{I}{4} + \frac{I}{2}$$

and defines the final visual notation:

$$X - X - Y = \frac{1}{4} - \frac{1}{4} - \frac{1}{2}.$$

- ## 1B. LINEAR BIDIRECTIONAL P.B.F



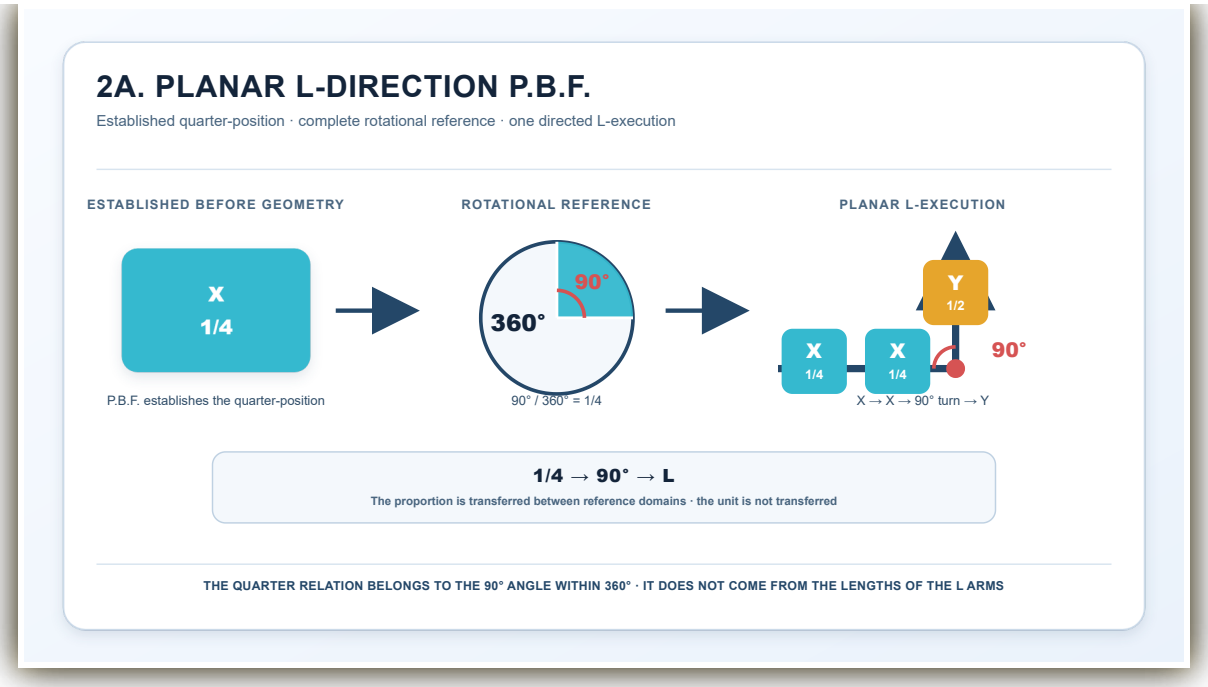
The image shows:

```
Y — X — X ← CENTER → X — X — Y
```

The sequence viewed from the center is `X–X–Y` in both directions. It also establishes that:

- the same P.B.F. function is used in both directions;
- the center is the common reference position;
- the direction changes, but the positional identity is preserved;
- the values of the branches are not to be added together;
- no third primary function is introduced.

- ## 2A. PLANAR L-DIRECTION P.B.F



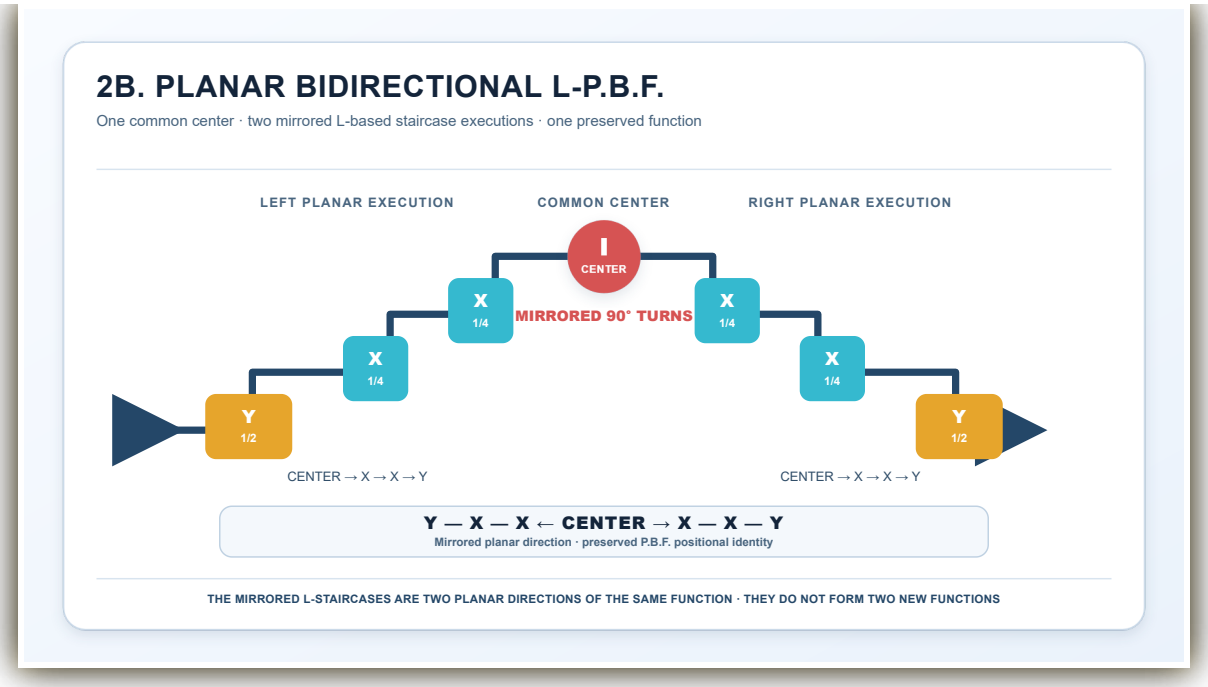
The image captures the transition:

$$\frac{1}{4} \longrightarrow 90^\circ \longrightarrow L$$

It explicitly shows that:

- P.B.F. establishes $\frac{1}{4}$ prior to the geometry;
- $\frac{90^\circ}{360^\circ} = \frac{1}{4}$;
- the L-shape represents this geometric quarter;
- `X–X–Y` continues through a directional 90° change;
- the quarter derives from the angle relative to the full rotation, not from the lengths of the L-arms.

- ## 2B. PLANAR BIDIRECTIONAL L-P.B.F



The image shows:

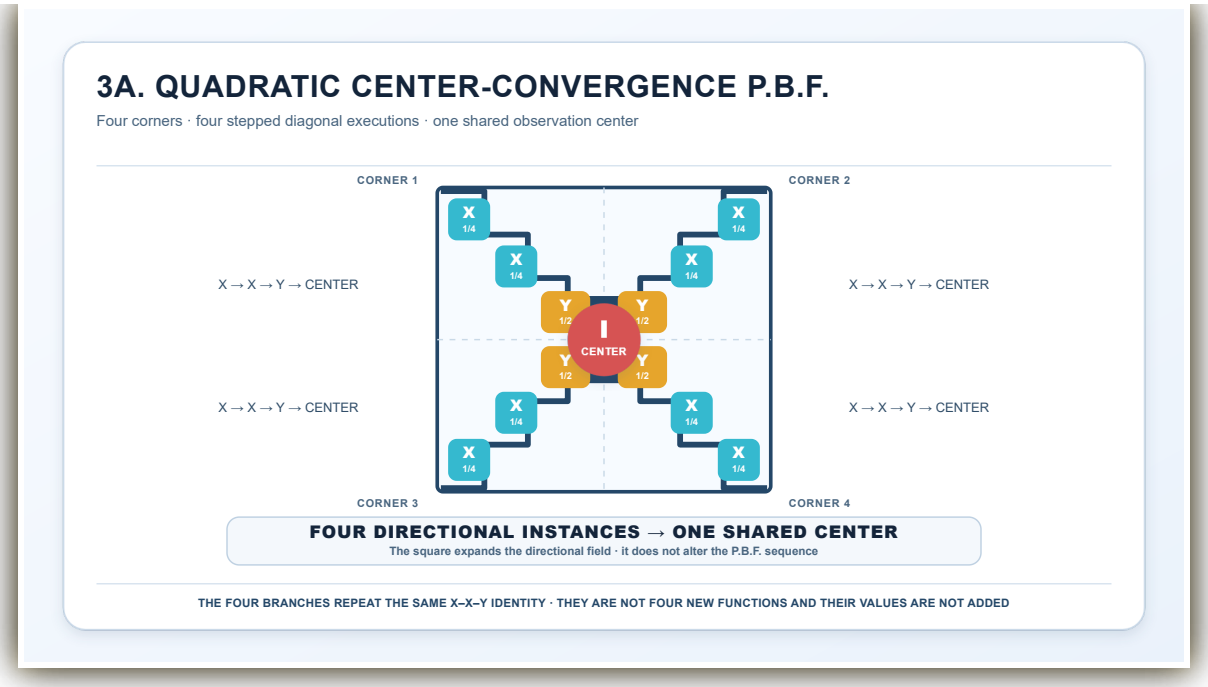
$$Y—X—X \leftarrow CENTER \rightarrow X—X—Y$$

formed by two mirrored L-based pyramid staircases.

It establishes that:

- both executions originate from the same center;
- each direction preserves the sequence `X–X–Y`;
- each step relies on perpendicular 90° changes;
- the mirroring alters direction but not functional identity;
- the branches are two planar execution paths of the same P.B.F. function;
- no additional primary functions are created.

- ## 3A. QUADRATIC CENTER-CONVERGENCE P.B.F



The image establishes:

```
FOUR CORNERS
→
FOUR STEPPED X–X–Y EXECUTIONS
→
ONE SHARED CENTER
```

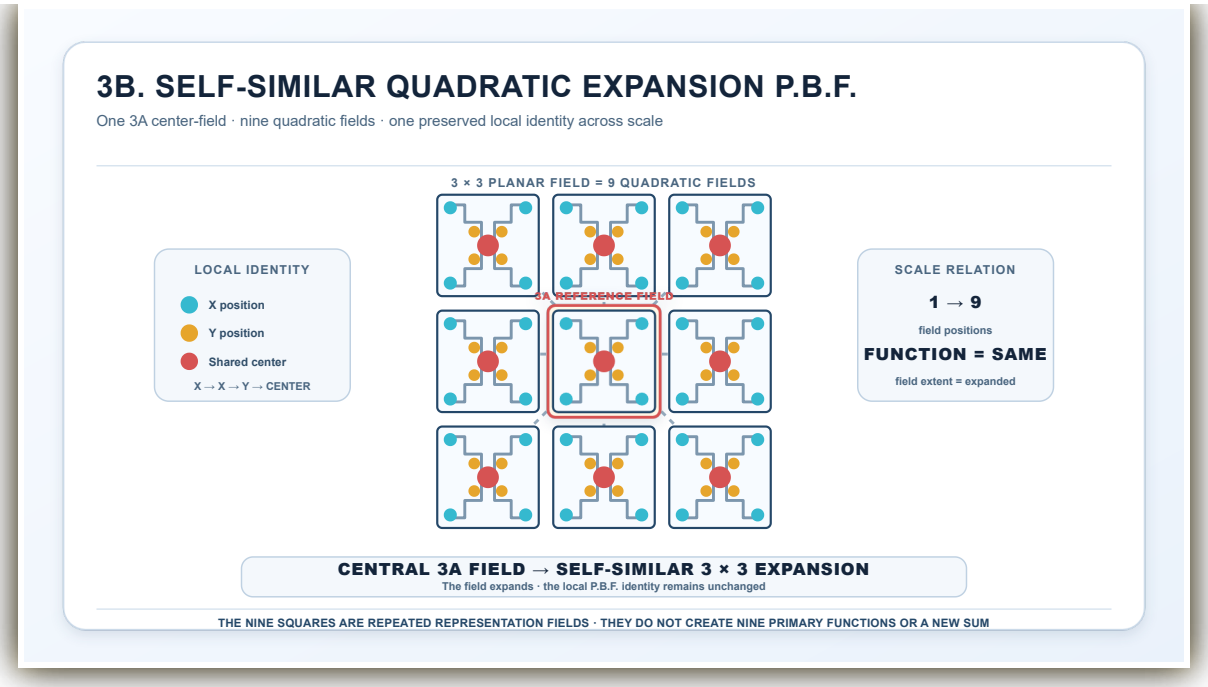
Each corner follows:

```
X → X → Y → CENTER
```

It establishes that:

- the square constitutes a shared planar directional field;
- all four branches retain the same P.B.F. positional identity;
- the stepped form consists of repeated perpendicular changes in direction;
- the center point is a shared observation and convergence position;
- the branches are not four new functions;
- their values are not to be added together.

- ## 3B. SELF-SIMILAR QUADRATIC EXPANSION P.B.F



The image shows:

$3 \times 3 = 9$

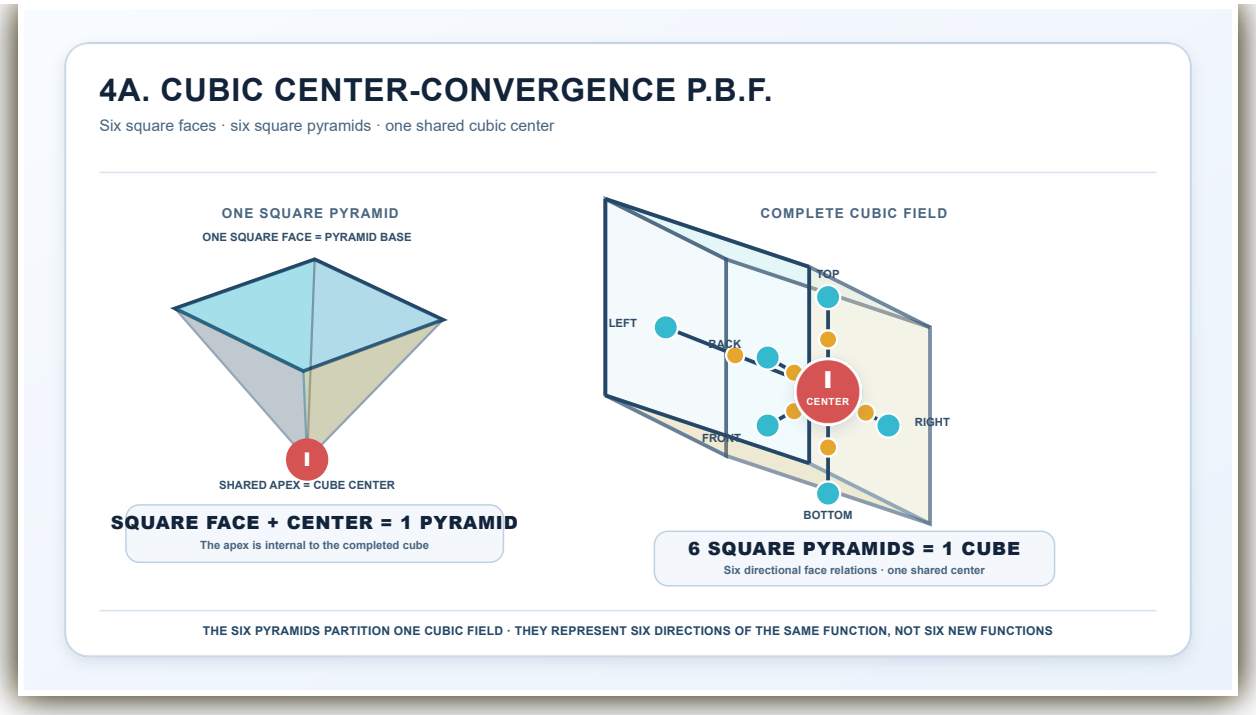
square representation fields, where the central square is the previous `3A` structure.

It establishes that:

- `3A` is preserved as the central reference field;
- the same local center convergence recurs in all nine squares;
- each local field preserves the relationship `X → X → Y → CENTER`;
- the scope of the field expands while the P.B.F. identity remains unchanged;
- the nine squares are self-similar representation fields;
- they do not create nine new primary functions or a new sum.

- CUBIC CENTER-CONVERGENCE P.B.F

4A.1. CUB



The image first establishes the geometric unit:

```
ONE SQUARE FACE
+
ONE INTERNAL CENTER
=
ONE SQUARE PYRAMID
```

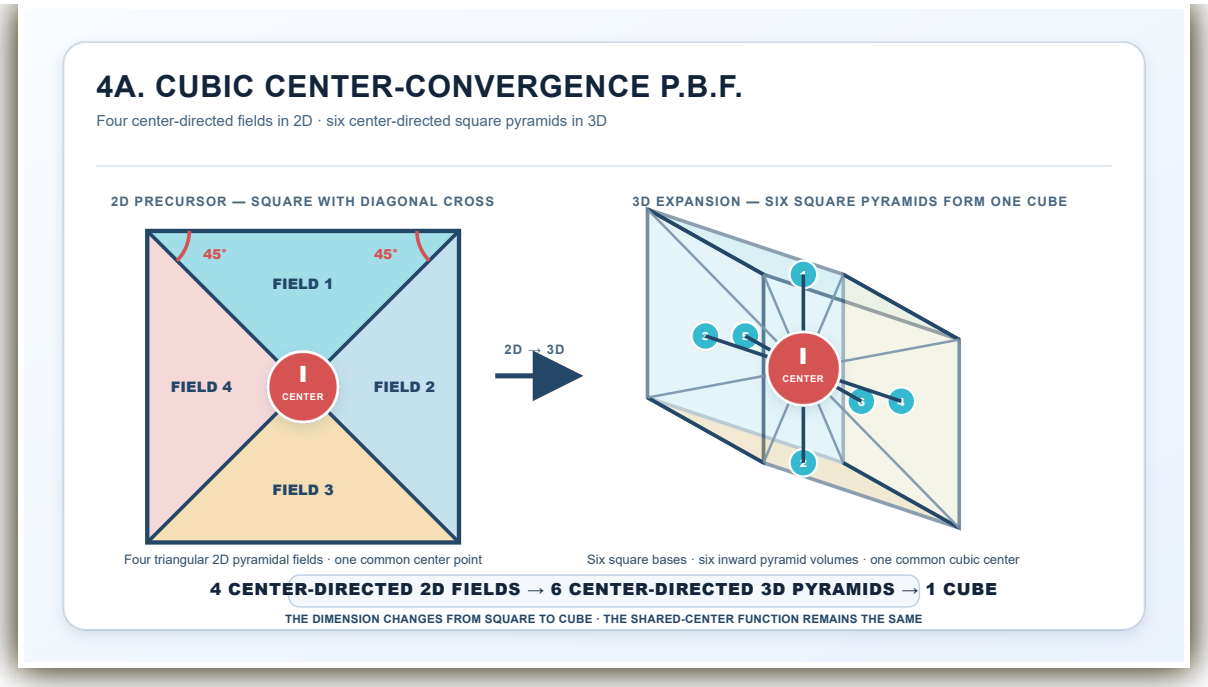
Next, the complete cubic relationship is shown:

$$6 \text{ square pyramids} = 1 \text{ cube}.$$

It establishes that:

- the cube's six faces serve as six pyramid bases;
- all the pyramids share the same internal center;
- the directions `TOP`, `BOTTOM`, `LEFT`, `RIGHT`, `FRONT`, and `BACK` converge toward the center;
- the six pyramids partition a single cubic field;
- they represent six directions of the same P.B.F. function;
- they do not form six new primary functions.

- ## 4A.2 CUBIC CENTER-CONVERGENCE P.B.F.



It first shows:

```
2D SQUARE
+
DIAGONAL CROSS
=
4 TRIANGULAR PYRAMIDAL SECTIONS
WITH A SHARED APEX AT THE CENTER
```

The diagonals are oriented at a 45° angle relative to the sides of the square.

Next, the dimensional transition is shown:

```
4 CENTER-ORIENTED 2D SECTIONS
→
6 CENTER-ORIENTED 3D PYRAMIDS
→
1 CUBE
```

In the cube:

- each face of the cube serves as the square base for a pyramid;
- each pyramid extends from its face to the center point of the cube;
- all six pyramid apices meet at the center;
- the six pyramid volumes together fill the cube;
- the same center-oriented function is preserved as the representation moves from 2D to 3D.

The previous version has been replaced, not duplicated. The next `4B` can now be correctly constructed from this structure as a $3 \times 3 \times 3$ array of 27 cubes.

- **The step-by-step progression:**

```
1 WHOLE CUBE
→
8 PARTS
→
16 PARTS
→
27 PARTS
```

Each step builds upon the cubic center structure already established in `4A`. Therefore, an illustrative scale suffices:

- a clearly drawn original cube;
- a cube divided into 8 parts;
- a simplified representation of 16 parts;
- a cube divided into 27 parts;
- a marking indicating that each part inherits the same local `4A` function.

$$2 \times 2 \times 2 = 8$$

and:

$$3 \times 3 \times 3 = 27.$$

```
1 CUBIC FIELD
→
8 CUBIC SUBFIELDS
→
16 REPRESENTATIONAL POSITIONS
→
27 CUBIC SUBFIELDS
```

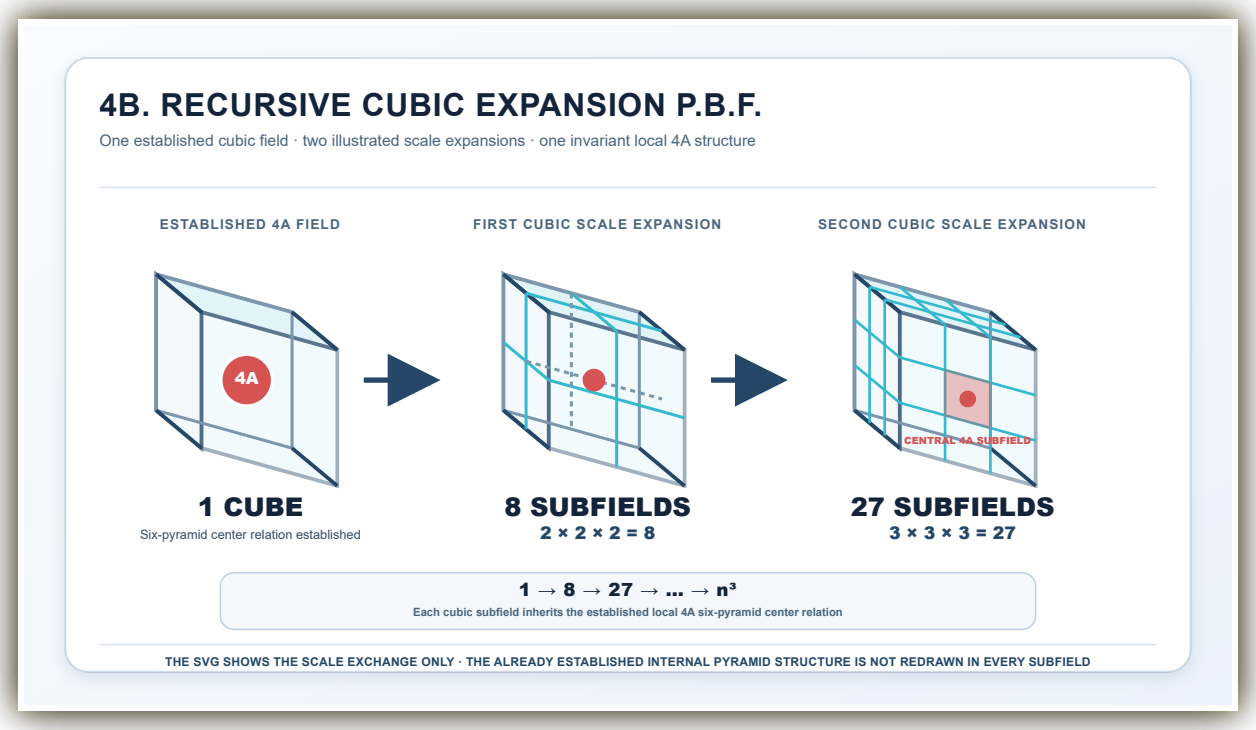
- **The regular cubic sequence is:**

$$1^3 = 1, \qquad 2^3 = 8, \qquad 3^3 = 27, \qquad 4^3 = 64, \ldots$$

The completed `4B` therefore shows only:

```
1 CUBE
→
8 CUBIC SUBFIELDS
→
27 CUBIC SUBFIELDS
→
n³
```

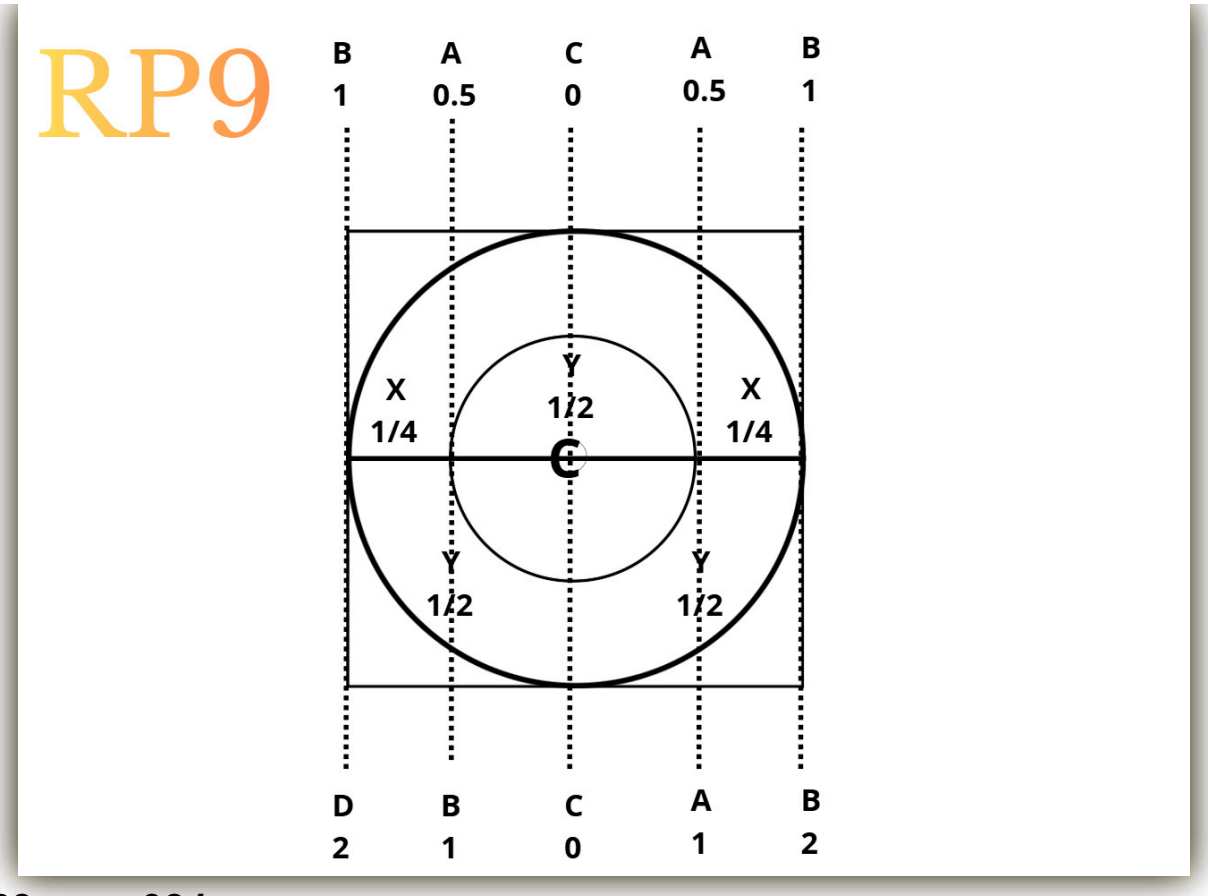
4B. RECURSIVE CUBIC EXPANSION P.B.F



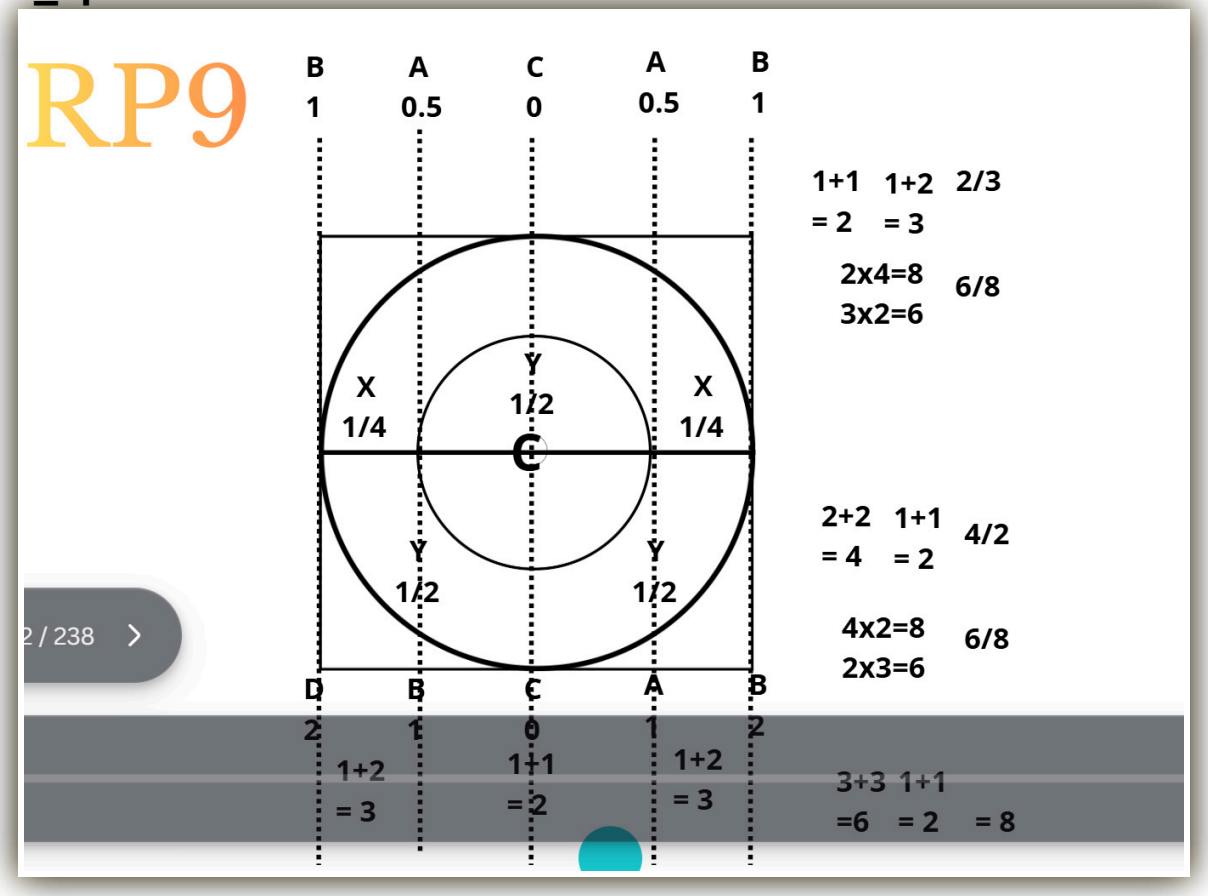
4B establishes that:

- `1` is the established `4A` cube;
- $2 \times 2 \times 2 = 8$ is the first illustrated cubic shell expansion;
- $3 \times 3 \times 3 = 27$ is the second;
- each sub-cube inherits the local six-pyramid structure from `4A`;
- the six pyramids therefore do not need to be redrawn within each sub-cube;
- continued regular expansion is generally expressed by n^3.

- **RP9_apx 090**



- **RP9_apx 091**



These two images introduce a new function not found in the previous SVG images.

They do not primarily show further geometric expansion. Instead, they demonstrate how multiple proportional relationships can be extracted simultaneously from the same original circle-square construction.

The first image establishes a clear coordinate and proportion field:

$$-1,\ -\tfrac{1}{2},\ 0,\ \tfrac{1}{2},\ 1$$

with the center at:

$$C = 0.$$

The same positions are expressed on the lower scale as:

$$-2,\ -1,\ 0,\ 1,\ 2.$$

These are two scale representations of the same symmetrical positions.

The horizontal diameter relationship can be read as:

$$\tfrac{1}{4} + \tfrac{1}{2} + \tfrac{1}{4} = 1.$$

Thus:

```
X – Y – X
1/4 – 1/2 – 1/4
```

Here, the values derive directly from the construction:

- the outer circle defines the full diameter;
- the inner circle defines the central half;
- the two remaining side sections each constitute a quarter;
- the center line and the parallel lines of observation make the relationships simultaneous.

The key point is that the image does not merely show one value at a time. The same construction simultaneously encompasses:

$$\tfrac{1}{4},\qquad \tfrac{1}{2},\qquad 1,\qquad 2$$

depending on which reference scale and relationship are being read. The second image then begins to show how the same positional field can be grouped using different orders of operations:

$$1+1=2,\qquad 1+2=3,\qquad 2\times 4=8,\qquad 3\times 2=6,$$

yielding:

$$\tfrac{6}{8} = \tfrac{3}{4}.$$

The mirrored grouping yields the same final relationship:

$$4 \times 2 = 8, \qquad 2 \times 3 = 6,$$

and thus, once again:

$$\frac{6}{8} = \frac{3}{4}.$$

This means that `6/8` does not need to be introduced from the outside. It can be shown as a dimensional and grouping representation of the already established relationship:

$$\frac{6}{8} = \frac{3}{4}.$$

The truly valuable aspect of the images is therefore:

```
A CONSTRUCTION
→
MULTIPLE SIMULTANEOUS READINGS
→
DIFFERENT SCALES
→
DIFFERENT GROUPING ORDERS
→
THE SAME RELATIONAL IDENTITY
```

$$1 \to 8 \to 27 \to n^3.$$

```
5A. CIRCULAR RELATIONAL EXTRACTION P.B.F.
```

First image:

```
CIRCLE–SQUARE ORIGIN
→
SYMMETRIC POSITION FIELD
→
1/4 – 1/2 – 1/4
→
TWO COMPATIBLE SCALES
```

and:

```
5B. EIGHT-PART RELATIONAL PROJECTION P.B.F.
```

Second image:

```
POSITIONAL GROUPING
→
2 × 4 = 8
→
3 × 2 = 6
→
6/8 = 3/4
```

It is also here that the connection to eight becomes comprehensible without needing to use eight as the origin of the construction. Eight emerges as a subsequent level of division and representation within a structure that has already established

$$14\frac{1}{4}, 12\frac{1}{2}, 34\frac{3}{4}$$

and the whole.

The material quite clearly alters the understanding of your sketches.

They show not only that multiple fractional values can be placed within a single figure.

They can demonstrate how `8` emerges as the first complete linear reference capable of simultaneously carrying and reproducing the relationships from the original circle-square construction.

The connecting chain within the material is:

```
R
↓
CIRCLE
↓
CONTINUOUS RADIAL RELATION
↓
MINIMAL ENCLOSING SQUARE
↓
4 × 2R
↓
8R
↓
Rx8
```

This means that `8` is not introduced because the number eight itself is assumed to be fundamental. It arises geometrically because:

```
CIRCLE DIAMETER = 2R

SQUARE SIDE = 2R

4 SQUARE SIDES × 2R = 8R
```

What `8` actually becomes

Your phrasing regarding `8` as an "extremity" is apt if "extremity" here signifies:

```
THE FIRST COMPLETE LINEAR ENCLOSURE
OF THE ORIGINAL RADIAL RELATION
```

Thus, Rx8 is not merely a final value. It becomes the complete external frame of reference within which the internal relations can be expressed simultaneously.

```
R
=
ORIGINATING REFERENCE

2R
=
COMPLETE DIAMETER

8R
=
COMPLETE LINEAR ENCLOSURE
```

This is precisely the aspect that my earlier interpretation of your sketches did not fully capture.

What your sketches can actually demonstrate

The first sketch can be constructed as the extraction field itself:

```
ONE CIRCLE–SQUARE CONSTRUCTION
↓
ONE INVARIANT CENTER
↓
ONE COMPLETE Rx8 REFERENCE
↓
MULTIPLE SIMULTANEOUS RELATIONAL READINGS
```

From this same construction, one can then derive:

$$\frac{1}{8}, \quad \frac{2}{8} = \frac{1}{4}, \quad \frac{3}{8}, \quad \frac{4}{8} = \frac{1}{2}, \quad \frac{6}{8} = \frac{3}{4}, \quad \frac{8}{8} = 1.$$

These are not six separate origins.

They are distinct relational positions within the same complete reference of eight. This is what makes the image interesting:

```
THE RELATIONSHIPS DO NOT CREATE Rx8 RETROACTIVELY.

Rx8 IS THE COMPLETE LINEAR REFERENCE
THROUGH WHICH THEY CAN BE DISPLAYED SIMULTANEOUSLY.
```

The connection to `3/3` and `4/4`

The accompanying material also introduces an important distinction in levels:

```
CIRCLE
=
3/3
=
RELATIONAL EQUILIBRIUM
```

```
SQUARE
=
4/4
=
COMPLETE LINEAR REPRESENTATION
```

and subsequently:

```
Rx8
=
COMPLETE LINEAR ENCAPSULATION
```

This implies that the images should distinguish between three things:

Level	Function
`3/3`	The complete circular relationship
`4/4`	The complete linear representation of the relationship

Level	Function
`8R`	The complete linear encapsulation calculated from `R`

These are not three competing wholes. They are three levels of expression for the same geometric origin.

The central relational transition

The S.4 material states:

```
RELATION
1/3 ⇔ 2/3

↓

REPRESENTATION
1/4 ⇔ 3/4
```

Your "8-reference" can then make the measurable representation more explicit:

$$\frac{1}{4} = \frac{2}{8}, \qquad \frac{3}{4} = \frac{6}{8}.$$

This yields:

```
CIRCULAR RELATION
1/3 ⇔ 2/3

↓

LINEAR REPRESENTATION
1/4 ⇔ 3/4

↓

Rx8 PROJECTION
2/8 ⇔ 6/8
```

This is far more precise than simply stating that:

$$\frac{6}{8} = \frac{3}{4}.$$

It shows where the representations belong and that they should not be confused with the origin of the relationship.

The true primary function of the images

I would therefore no longer label them solely as:

```
EIGHT-PART RELATIONAL PROJECTION
```

A more precise structure is:

```
5A. Rx8 RELATIONAL REFERENCE FIELD
```

It shows:

```
R
→
CIRCLE
→
MINIMAL SQUARE
→
8R
→
COMPLETE RELATIONAL REFERENCE FIELD
```

And subsequently:

```
5B. SIMULTANEOUS RELATIONAL EXTRACTION THROUGH Rx8
```

It shows, for example:

```
1/8
2/8 = 1/4
3/8
4/8 = 1/2
6/8 = 3/4
8/8 = 1
```

as well as how the sequence of multiple groupings can yield the same preserved relationship.

The defining declaration

```
Rx8 IS NOT AN ARBITRARY DIVISION INTO EIGHT PARTS.

Rx8 IS THE FIRST COMPLETE LINEAR ENCAPSULATION
GENERATED FROM THE INVARIANT RADIAL REFERENCE R.

THE EIGHT-PART REFERENCE DOES NOT CREATE
THE RELATIONS CONTAINED WITHIN THE GEOMETRY.

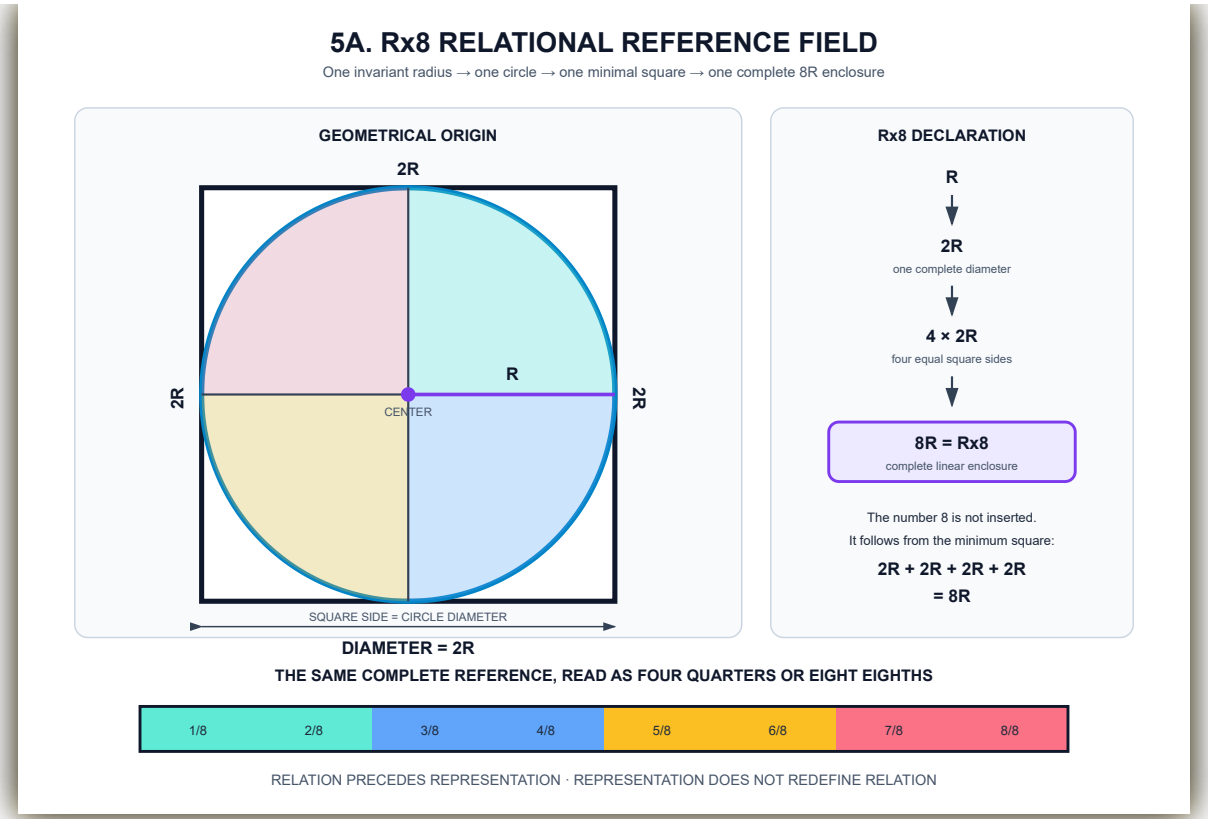
IT PROVIDES THE COMPLETE LINEAR FIELD THROUGH WHICH
THESE RELATIONS CAN BE POSITIONED, COMPARED,
AND OBSERVED SIMULTANEOUSLY.
```

In doing so, they reveal more than just "parts of eight." They reveal:

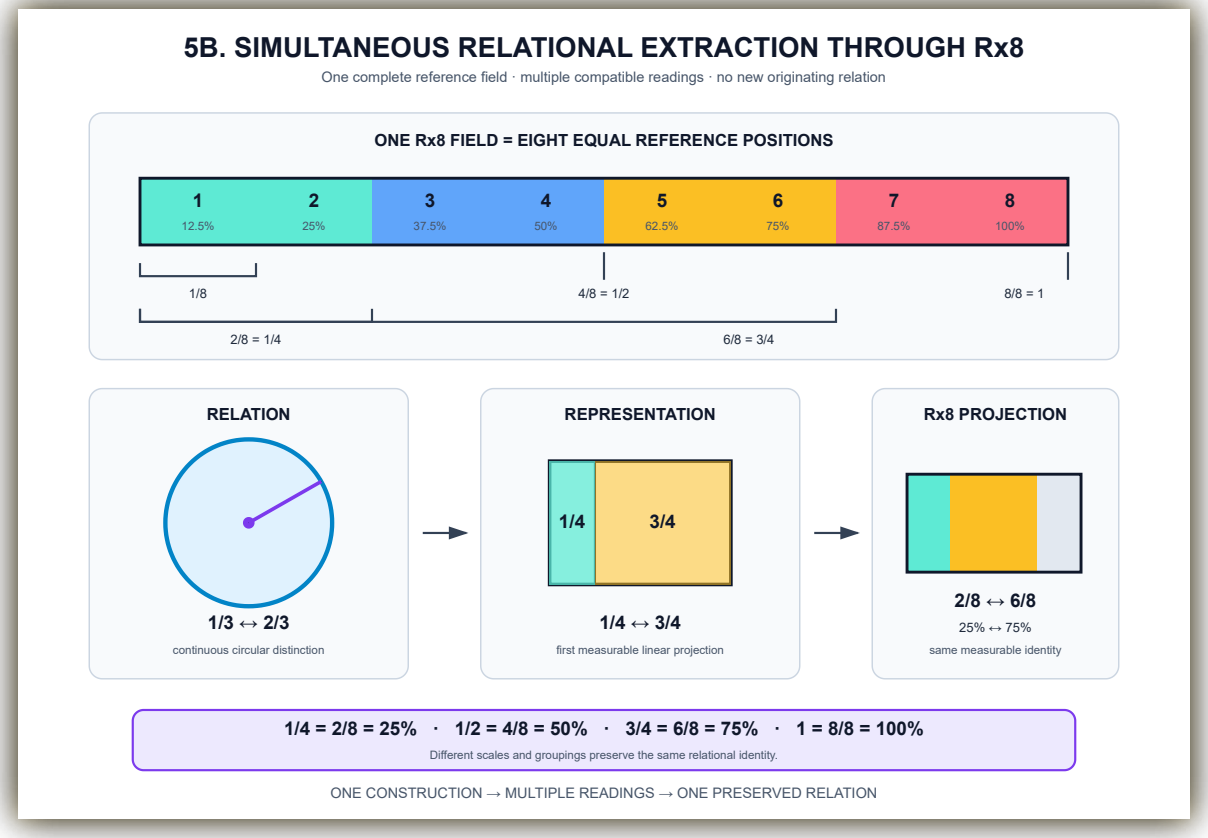
```
A GEOMETRIC ORIGIN
→
A COMPLETE LINEAR REFERENCE
→
MULTIPLE SIMULTANEOUS RELATIONS
→
THE SAME PRESERVED IDENTITY
```

That is where your sketches and the Rx8 declaration actually converge.

- ## 5A. Rx8 RELATIONAL REFERENCE FIELD



- **5B. SIMULTANEOUS RELATIONAL EXTRACTION THROUGH Rx8**



`5A` shows the geometric construction:

$$R \to 2R \to 4 \times 2R \to 8R = \mathrm{R8}$$

including the circle, the minimal enclosing square, the center, the radius, and the four equivalent quarter-sections.

`5B` shows the simultaneous readings:

$$\frac{1}{4} = \frac{2}{8} = 25$$

$$\frac{1}{2} = \frac{4}{8} = 50$$

$$\frac{3}{4} = \frac{6}{8} = 75$$

$$1 = \frac{8}{8} = 100.$$

It also shows the structural level transition:

```
RELATION
1/3 ⇔ 2/3

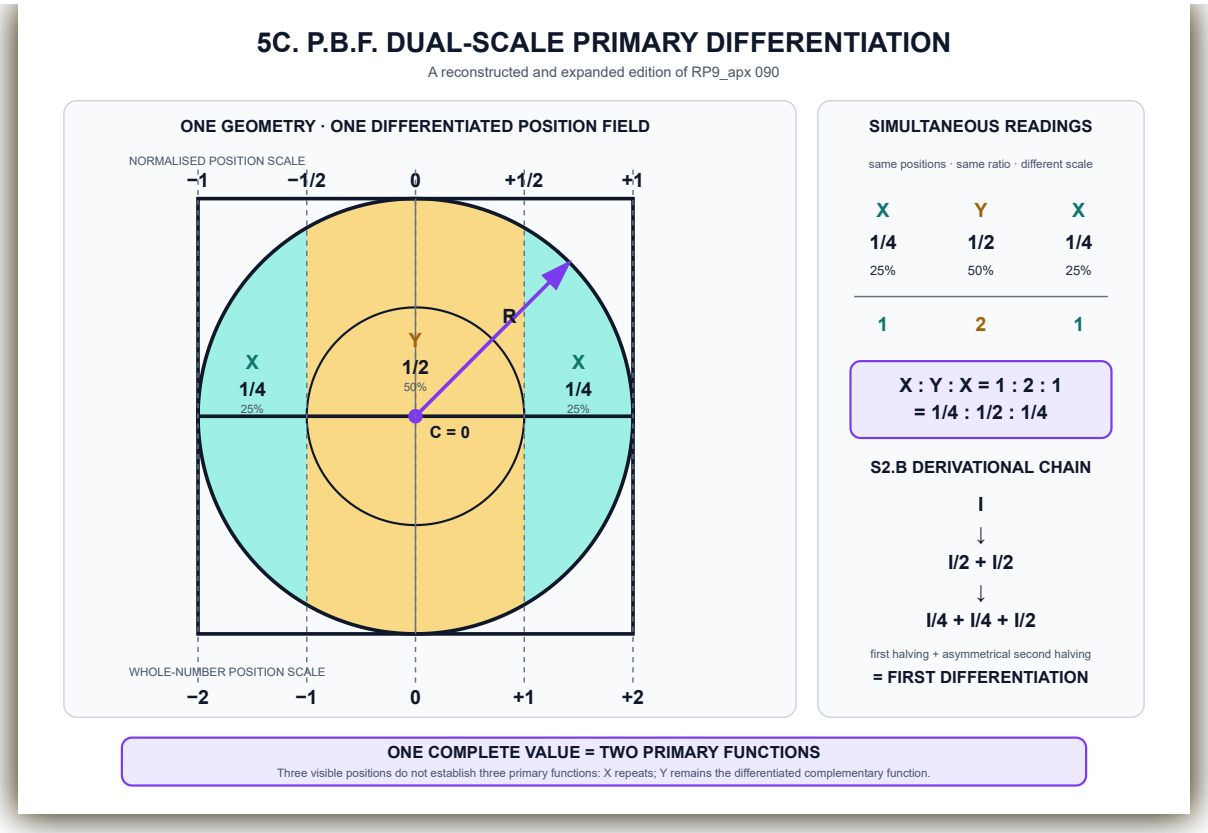
↓

REPRESENTATION
1/4 ⇔ 3/4

↓

Rx8 PROJECTION
2/8 ⇔ 6/8
```

- ## 5C. P.B.F. DUAL-SCALE PRIMARY DIFFERENTIATION



It shows the same positional structure through three simultaneous readings:

$$X - Y - X$$

$$\frac{1}{4} - \frac{1}{2} - \frac{1}{4}$$

$$1 - 2 - 1$$

with the identity:

$$X : Y : X = 1 : 2 : 1 = \frac{1}{4} : \frac{1}{2} : \frac{1}{4}.$$

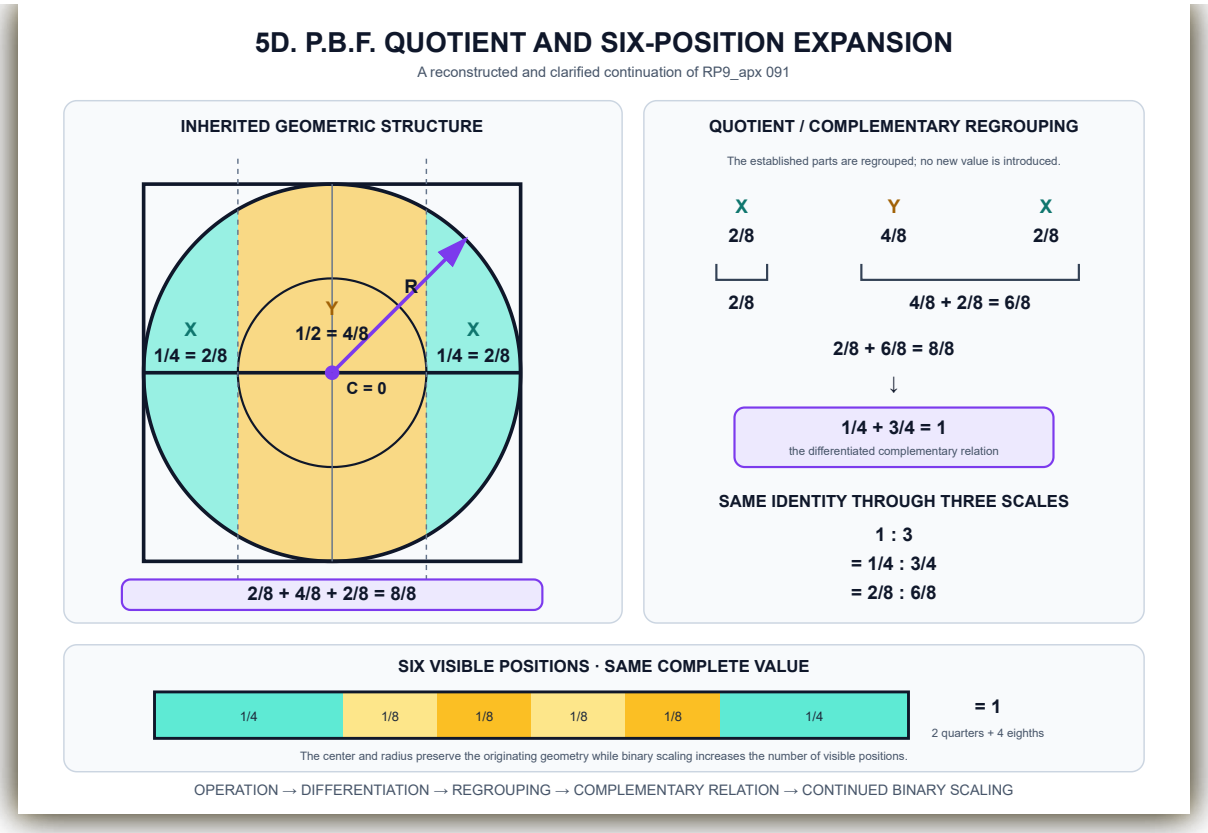
It also includes the repository's derivation chain:

$$I \longrightarrow \frac{I}{2} + \frac{I}{2} \longrightarrow \frac{I}{4} + \frac{I}{4} + \frac{I}{2}.$$

The color coding makes the functional difference immediately visible:

- both `X` positions have the same turquoise color;
- `Y` has a distinct yellow color;
- the center `C = 0` is marked in violet;
- both the normalized scale `−1 → 1` and the integer scale `−2 → 2` are preserved.

- ## 5D. P.B.F. QUOTIENT AND SIX-POSITION EXPANSION



`5C`, based on `RP9_apx 090`, shows:

$$X - Y - X$$

$$\frac{1}{4} - \frac{1}{2} - \frac{1}{4}$$

$$1 - 2 - 1.$$

The circle, center `C = 0`, radius `R`, the enclosing square, and the two simultaneous scales are preserved. `X` is rendered in the same color on both sides, and `Y` as the differentiated middle function.

`5D`, based on `RP9_apx 091`, continues from the same geometry and shows:

$$\frac{2}{8} + \frac{4}{8} + \frac{2}{8} = \frac{8}{8}$$

as well as the regrouping:

$$\frac{2}{8} + \left(\frac{4}{8} + \frac{2}{8}\right) = \frac{2}{8} + \frac{6}{8} = \frac{8}{8},$$

which corresponds to:

$$\frac{1}{4} + \frac{3}{4} = 1.$$

It also shows the scale identity:

$$1:3=\frac{1}{4}:\frac{3}{4}=\frac{2}{8}:\frac{6}{8}.$$

At the bottom is the six-position expansion of S2.B:

$$\frac{1}{4}+\frac{1}{8}+\frac{1}{8}+\frac{1}{8}+\frac{1}{8}+\frac{1}{4}=1.$$

The key point is that `5D` does not depart from the geometric foundation. The circle, center, and radius remain, while the fractional division and the continued binary scaling are superimposed on the same underlying structure.

NEXT LEVEL: RX8

R8.A

“RP9. RH. Z0/000. RP9. EXTRA ACCESSORIES (RH. Z0)/C. P.B.F. EXPANSION/T3/VISUAL EXPANTION P.B.F/P.B.F. SVG/R8.A.png” kunde inte hittas.

The Geometrical Origin of the Rx8 Reconstruction

R8.A introduces the fundamental geometrical construction from which the Rx8 reconstruction originates.

Beginning with a single invariant radius **R**, the circle is enclosed by its minimal square whose side length equals the circle's diameter, **2R**. The resulting perimeter therefore becomes **8R**, establishing the first complete linear encapsulation generated directly from the original radial reference.

The significance of this construction does not lie in the numerical value itself.

Its significance lies in the transition from continuous circular geometry to the first complete linear reference framework capable of supporting differentiation, measurement, relational reconstruction, and recursive propagation.

Within the RP9 framework, **R8.A** serves as the geometrical foundation upon which the complete Rx8 reconstruction is developed.

R8.B

“RP9. RH. Z0/000. RP9. EXTRA ACCESSORIES (RH. Z0)/C. P.B.F. EXPANSION/T3/VISUAL EXPANTION P.B.F/P.B.F. SVG/R8.B.png” kunde inte hittas.

Architectural Overview of the Rx8 Reconstruction

R8.B provides a condensed architectural overview of the complete Rx8 reconstruction.

Rather than presenting the full geometrical derivation, it summarizes the principal concepts, operational functions, structural relationships, and sequential development that define the Rx8 framework.

Beginning with the original geometrical reference **R**, the overview follows the progression toward the first linear encapsulation **Rx8**, establishing the methodological foundation for relational differentiation, structural representation, recursive reconstruction, and the subsequent verification architecture developed throughout RP9.

Within the repository, **R8.B** functions as the architectural introduction to the complete Rx8 documentation, providing the conceptual context required for the detailed declarations and verification presented in the accompanying documents.
